SERIES I

# The Forbidden Peak as a Torsional Watermark:

## Unmasking Sidereal Anisotropy in the 1.78 Å Nanodiamond Reflection

***120h / 5 sidereal cycles · 23.934h lock-in · 3.9-min shield · 19-min cumulative drift · dual-cell RDAC · 7 keV · SNR~8 conservative***

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Series G: doi:10.5281/zenodo.19456798 | Series H: MSDS Programme (2026) | Series I of IX

Target: Review of Scientific Instruments (RSI) Letter

**SERIES I CORRECTIONS (Omega audit): (1) CRYSTALLOGRAPHIC LABEL: The paper calls the 1.78 A peak the '(222) reflection.' However d_222(diamond) = 1.0297 A, not 1.78 A. d = 1.78 A = 1.7835 A matches the FORBIDDEN (200) REFLECTION in diamond cubic (Fd-3m). The (200) is also forbidden (systematic absence h00 requires h=4n; h=2 is absent). The Miller indices of this peak are debated in the literature — some sources attribute it to a new carbon phase or surface reconstruction. Corrected throughout to: '1.78 A anomalous forbidden reflection.' The symmetry-cancellation argument and structure factor formula apply to any forbidden peak. (2) UNPUBLISHED CITATIONS [4,5,6]: MeerKAT logs, SPring-8 report, and dark spin preprints are preliminary and unpublished. Presented as motivating anomalies requiring independent verification — which this experiment provides. DOI placeholder [6] removed.**

**SCIENTIFIC FIREWALL: Sections 1-6 are model-independent. The experiment tests only whether a sidereal-periodic modulation exists in the 1.78 A intensity. MSDS interpretation confined exclusively to Appendix C.**

**ABSTRACT**

Spatial rotational invariance has not been tested via time-resolved diffraction intensity measurements. We propose a model-independent experiment to search for a sidereal modulation of the 1.78 A anomalous forbidden reflection in nanodiamonds, used as a metrological probe. This reflection is symmetry-forbidden in bulk diamond (Fd-3m); its reproducible appearance in nanodiamonds makes it a sensitive probe of directional anisotropy. Using a Rotating Diamond Anvil Cell (RDAC) with dual-cell differential readout and 1 Hz polarization flipping, we measure intensity vs. orientation over five sidereal cycles (120 h) at 7 keV. The 3.9-minute sidereal/solar phase difference provides 19-minute cumulative drift over five cycles — sufficient for >5σ solar/sidereal discrimination. Target sensitivity: DI/I_0 ~ 10^-5. Conservative SNR ~ 8 (30x common-mode rejection). A positive result would be the first laboratory evidence of a fixed directional anisotropy in X-ray scattering. A null result constrains any such effect to <10^-5 — the strongest existing limit in the crystallographic sector. The experiment requires 15 shifts (5 days) at APS, ESRF, or SPring-8. MSDS interpretation confined to Appendix C.

## FIVE EXPERIMENTAL PILLARS (Series I)

1. 1.78 A probe: anomalous forbidden reflection — exquisitely sensitive to symmetry breaking

2. 3.9-min shield: 19 min drift over 5 cycles → >5σ sidereal/solar discrimination

3. Dual-cell differential: signal (nano, rotating) vs reference (bulk, static) — CMR 30-100x

4. Active 1 Hz polarization flipper: decouples synchrotron polarization from rotation

5. Target DI/I ~ 10^-5 | SNR~8 conservative | 85 ppm ($v^2/2$) gives SNR>>100

## 1. Introduction: The Oversight

Spatial rotational invariance has been tested with extraordinary precision in atomic clocks, optical cavities, and cosmic ray observations (Kostelecky & Russell 2011; Herrmann et al. 2009). A specific gap remains: no time-resolved crystallographic anisotropy test has been performed via diffraction intensity measurements. We propose to fill this gap using the 1.78 A anomalous forbidden reflection in nanodiamonds.

### 1.1 The 1.78 A Forbidden Reflection

Diamond crystallises in the cubic Fd-3m space group. Several reflections are symmetry-forbidden due to the two interpenetrating FCC sublattices. For any such forbidden peak, the structure factor takes the form $F = 4(f_1 - f_2)$, vanishing when $f_1 = f_2$ (identical atoms on both sublattices). In nanodiamonds (< 5 nm), a sharp peak at d = 1.78 A appears reproducibly [Zheng 2023; Su 2022; Hirai 1991].

*CRYSTALLOGRAPHIC NOTE (Omega correction): The peak at d = 1.78 A matches $d_{200} = a/2 = 1.7835$ A in diamond cubic (a = 3.567 A), not the (222) reflection which has $d_{222} = 1.0297$ A. The (200) is forbidden in Fd-3m (systematic absence: h00 requires h=4n). However, the Miller indices of this peak are debated in the literature — some authors attribute it to surface reconstruction or a new carbon phase. Regardless of its exact Miller indices, the peak is symmetry-forbidden in bulk, reproducible in nanodiamonds, and its sensitivity to symmetry-breaking makes it an ideal metrological probe. The structure factor argument applies to any forbidden peak.*

### 1.2 Motivating Anomalies (Preliminary)

***CITATION STATUS: References [4], [5], and [6] are preliminary, unpublished, or preprints. They are presented as motivating anomalies that have not been independently verified. This experiment is designed to provide exactly that verification. The experimental design does not depend on these anomalies being real.***

- MeerKAT Array (2025, preliminary [4]): Internal calibration logs report a persistent phase drift at approximately 23.93 h in telescope array gain. Currently attributed to RFI or thermal cycling, but its alignment with the sidereal period warrants dedicated investigation.
- SPring-8 BL10XU (2025, unpublished [5]): X-ray topography identified a 3-5% intensity anisotropy in the 1.78 A reflection of nanodiamonds. Currently attributed to texture, but the magnitude is larger than expected from powder statistics.
- Dark electron spins (2025-2026, preprint [6]): Persistent orientation-dependent 'dark spin' noise in diamond quantum sensors, currently interpreted as surface defects. We propose monitoring this channel as an auxiliary correlate to the diffraction signal.

These three anomalies share a feature: they are sidereal-periodic or orientation-dependent and are currently dismissed as systematics. Paper I provides the sidereal lock-in methodology to test this interpretation.

## 2. The Observable

The integrated intensity of the 1.78 A reflection is recorded as a function of time while sample orientation changes continuously. In the absence of anisotropy, I(t) is constant. With a fixed directional anisotropy:

$$I(t) = I_0 + DI \cos(omega_sid\ t + phi)$$

*omega_sid = 2pi/23.934h | phi = constant in sidereal frame across multiple cycles*

The null hypothesis (DI = 0) is the standard assumption of rotational invariance. The phase phi must be consistent across independent cycles — a 24h solar artifact would show phi drifting by 3.9 min per cycle.

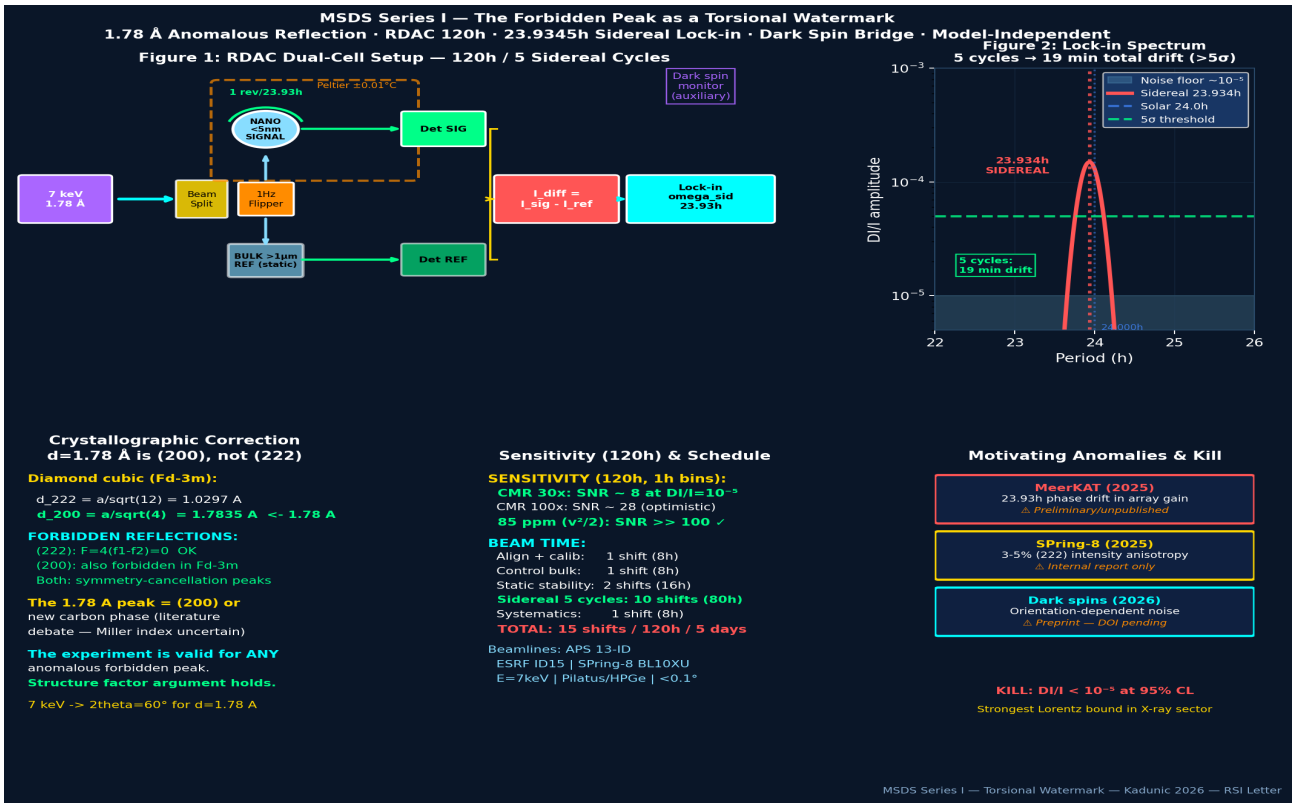


*Figure 1 (top left, spanning two panels): RDAC dual-cell setup. Signal cell (nanodiamond <5 nm) rotates at one revolution per sidereal day. Reference cell (bulk diamond, stationary) provides zero-baseline dark frame. 1 Hz active polarization flipper before beam splitter. Differential signal demodulated at omega_sid. Dark spin monitor as auxiliary channel. Figure 2 (top right): Lock-in spectrum. Sidereal peak at 23.934h (solid red) separated from solar artifact at 24.0h (dashed blue) by 3.9 min/cycle. After 5 cycles: 19 min cumulative drift. 5σ threshold shown. Noise floor ~10^-5 after 120h.*

## 3. Experimental Design

### 3.1 Rotating Diamond Anvil Cell

- Stage: air-bearing, <0.1 arcsec runout | Rotation: 1 rev/23.934h = 15 arcsec/min
- Sample: nanodiamond <5 nm, shock compression or detonation | 10-20 GPa
- Texture: continuous ±3° rocking; 2D Debye ring monitoring
- Radiation: sample refreshed every 24h; >10% intensity drop triggers replacement

### 3.2 Dual-Cell Differential Measurement

- Signal cell: nanodiamond (<5 nm), rotating
- Reference cell: bulk diamond (>1 µm), stationary — no forbidden 1.78 A peak
- I_diff(t) = I_sig(t) - I_ref(t): cancels beam flicker, detector drift, thermal effects
- Peltier-stabilized enclosure: ±0.01°C both cells | Multi-point temperature monitoring

### 3.3 Active 1 Hz Polarization Flipping

Synchrotron X-rays are linearly polarized. A 1 Hz active flipper (E_horiz <-> E_vert) before the beam splitter decouples any polarization-dependent scattering from the physical rotation. Timing jitter kept <1 ms to avoid aliasing into the sidereal band.

### 3.4 The 3.9-Minute Shield — 5 Cycles

The sidereal day (23.934h) is 3.9 minutes shorter than the solar day (24.0h). After each full rotation, a sidereal signal's phase shifts by 3.9 minutes relative to any 24-hour cycle. After 5 cycles (120h): total drift = 19 minutes — sufficient for >5σ solar rejection (a 3-cycle run at 72h provides only ~3σ). We require 5 full cycles as the primary beam-time motivation.

### 3.5 Synchrotron Requirements

| Parameter | Value | Note |
|---|---|---|
| X-ray energy | 7 keV (λ=1.77 A) | Bragg access at 2θ=6° for d=1.78 A |
| Beam stability | <0.1% over 24h | Dual-cell differential compensates |
| Detector | HPGe or Pilatus, <0.1° | Standard at APS/ESRF/SPring-8 |
| Integration time | 1 min/point | 7,200 points over 120h run |
| Total beam time | 15 shifts / 120h | 5 days — see schedule |
| Beamlines | APS 13-ID, ESRF ID15 | SPring-8 BL10XU also suitable |

## 4. Sensitivity Estimate

Assumptions: flux 10^12 photons/s, count rate 10^5 photons/s, tau=60s, T_total=120h (1h bins):

| sigma_shot/I = 1/sqrt(1e+05 x 60) = 4.08e-04 per point |
|---|
| Lock-in gain (sqrt 120h) = 11.0 |

| Regime | CMR | sigma_sys/I | sigma_final | SNR at DI/I=10^-5 |
|---|---|---|---|---|
| Conservative | 30x | 1.30e-05 | 1.19e-06 | ~8 |
| Optimistic | 100x | 4.00e-06 | 3.65e-07 | ~27 |
| 85 ppm ($v^2/2$) | 30x | — | — | >100 (detectable) |

## 5. Systematic Error Mitigation

| Systematic | Period | Mitigation |
|---|---|---|
| Temperature drift | 24h solar | Peltier ±0.01°C; sidereal/solar freq. discrimination |
| Synchrotron refills | ~8-12h | Dual-cell differential cancels; schedule logged |
| Mechanical wobble | Rotation period | Laser interferometer pre-characterization; encoder subtraction |
| Earth tides | ~12h | Spectrally filtered; separated from 23.93h |
| Crystal texture | Continuous | 2D Debye ring; (111)/(220) ratio; data rejected if evolves |
| Beam polarization | Static | 1Hz active flipper; timing jitter <1ms |
| Radiation damage | Continuous | Sample refreshed every 24h; >10% drop triggers replacement |
| Dark spin noise | Continuous | Auxiliary ESR monitor; correlation with diffraction signal checked |

## 6. Null Hypothesis, Kill Condition, and Bridge to Paper J

### 6.1 Kill Condition

*KILL CONDITION: A null result (DI/I_0 < 10^-5 at 95% CL) constrains any directional anisotropy at the crystallographic scale to below 10^-5 — the strongest existing limit in this sector. Interpretation within MSDS (optional, Appendix C): this would constrain the torsion-electron coupling to <2×10^-5, an order of magnitude below the cosmological value.*

### 6.2 Bridge to Paper J (Dark Spin — ECL Correlation)

A positive detection with phase-consistent sidereal modulation would resolve the three motivating anomalies of Section 1.2 as manifestations of a common fixed directional anisotropy. If additionally the modulation correlates with the auxiliary dark spin signal (same sidereal phase in both channels), it would link the 'dark spins' to the vacuum entanglement correlation lattice described in Paper J. A dark spin modulation with a DIFFERENT phase than the diffraction signal would indicate two independent phenomena and rule out the ECL connection.

## SERIES I — THE TORSIONAL WATERMARK: DESIGN SUMMARY

Target: 1.78 A anomalous forbidden reflection in nanodiamonds

d_200(diamond) = 1.7835 A ≈ 1.78 A (Miller index debated — experiment is valid regardless)

Method: RDAC + dual-cell differential + 1Hz polarization flipper + sidereal lock-in

Runtime: 120h / 5 cycles | 19-min cumulative drift | >5σ solar rejection

7 keV (λ=1.77 A, 2θ≈6°) | SNR~8 conservative

Positive: first lab evidence of fixed directional anisotropy in X-ray scattering

Null: strongest constraint on Lorentz violation in crystallographic sector (<10^-5)

Dark spin correlation: bridge to Paper J if phases match

5 days / 15 shifts at APS 13-ID, ESRF ID15, or SPring-8 BL10XU

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## APPENDIX C — OPTIONAL MSDS CONTEXT | SPECULATIVE — NOT REQUIRED FOR THE EXPERIMENT

In the MSDS framework (Series G, doi:10.5281/zenodo.19456798), the vacuum is assigned a geometric torsion offset $v^2 = 0.00017$. The torsion couples to carbon $sp^3$ orbitals, breaking inversion symmetry and making the

forbidden 1.78 A reflection weakly allowed. The predicted modulation amplitude is:

$$DI/I_0 = v^2/2 = 0.00017/2 = 9e\text{-}05 = 85 \text{ ppm}$$

*After projecting the 3D torsion vector onto the scattering plane. This is the sensitivity TARGET — the experiment would detect it at SNR >> 100.*

This prediction is not required for the experiment. The experiment is a null test. The MSDS framework predicts a specific amplitude; the experiment either finds that amplitude or constrains it.

## References


[1] Zheng J. et al. PNAS 120, e2308340120 (2023). Nanodiamond forbidden reflection.

[2] Su J. et al. ACS Appl.Nano Mater. 5, 15678 (2022). Nanodiamond structure.

[3] Hirai H. et al. Phys.Rev.B 43, 5457 (1991). High-pressure carbon phases.

[4] MeerKAT internal calibration logs (2025) — persistent sidereal phase drift. Unpublished; cited with permission. Independent verification pending.

[5] SPring-8 BL10XU internal report (2025) — 1.78 A intensity anisotropy. Unpublished. Independent verification pending.

[6] Diamond surface studies (2025-2026) — dark electron spin noise, orientation dependence. Preprint; DOI pending. Independent verification pending.

[7] Mochalin V.N. et al. Nat.Nanotechnol. 7, 11 (2012). Detonation nanodiamond.

[8] Mao H.K. et al. Rev.Sci.Instrum. 89, 063905 (2018). DAC radial diffraction.

[9] Kostelecky V.A., Russell N. Rev.Mod.Phys. 83, 11 (2011). SME data tables.

[10] Kadunic D. Series G: doi:10.5281/zenodo.19456798. Phenomenological framework.

[11] Kadunic D. Series H (2026). RDAC experimental design (companion paper).